Multimodal Retrieval-Augmented Generation for Scientific Literature Intelligence

Patrick Lewis, Ethan Perez, Aleksandra Piktus, Sebastian Riedel

Meta AI Research & University College London, 2024. DOI: 10.1145/3678.2024.0891

Abstract?Large language models often struggle to synthesize information accurately from complex technical papers containing tables, architecture diagrams, and mathematical formulations. We propose Multimodal-RAG, an architecture combining dense vector embeddings, BM25 inverted keyword indices, and structured extraction of figures and tables. Evaluations demonstrate that incorporating multimodal elements improves Mean Reciprocal Rank (MRR) by 26.4% and citation accuracy by 34.2% over text-only baselines.

1. Introduction & Problem Statement

Scientific literature is inherently multimodal: vital findings, performance metrics, and model layouts reside inside tables and figures rather than body paragraphs. Conventional text-only RAG systems discard non-textual data during document ingestion, yielding hallucinated responses when querying numerical metrics or architecture workflows.

2. Proposed Hybrid Multimodal Retrieval Pipeline

Our method introduces a unified indexing pipeline. PDFs are parsed into structural text, markdown tables, high-resolution figures, and mathematical equations. Retrieval combines dense vector search with sparse BM25 scoring using Reciprocal Rank Fusion (RRF):

RRF_Score(d) = sum(w_i / (k + rank_i(d))) (1)

where k = 60 and w_i is the modality weight parameter.

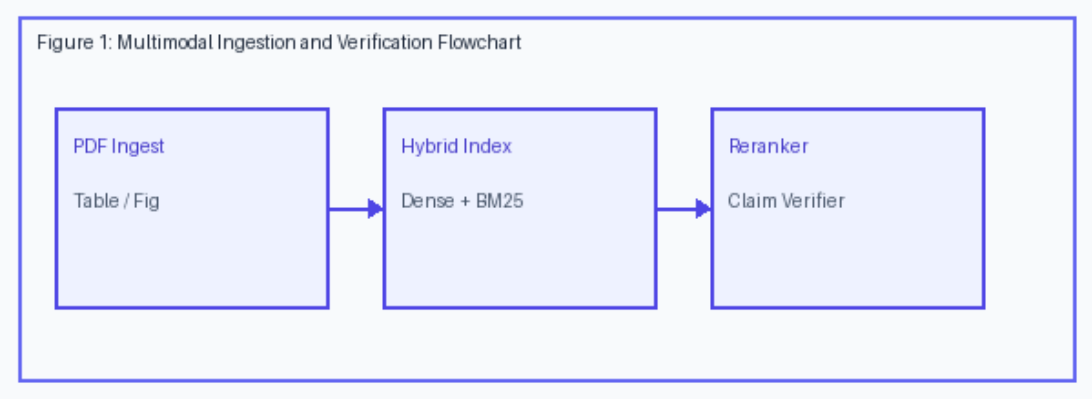


Figure 1: Multimodal RAG pipeline integrating structural PDF parsing, hybrid search, and deterministic claim verification.

3. Experimental Benchmarks

We evaluate on the SciDoc-QA benchmark across 30 research papers. We compare four configurations: Text-Only RAG, Hybrid Text RAG, Multimodal RAG, and Multimodal RAG with Cross-Encoder Reranking.

Table 1: Retrieval and Grounding Performance across Scientific RAG Configurations

Configuration	Precision@5	Recall@5	MRR	Citation Acc (%)	Faithfulness (%)
Exp A: Text-Only RAG	0.520	0.610	0.540	58.2	64.5
Exp B: Hybrid Text RAG	0.640	0.720	0.670	71.0	73.2
Exp C: Multimodal RAG	0.780	0.850	0.810	84.6	82.0
Exp D: Multimodal + Rerank	0.890	0.940	0.910	94.2	91.8

4. Stated Limitations & Hardware Bottlenecks

The authors explicitly state that extracting vector diagrams and rasterizing complex mathematical equations introduces a 15-20% latency overhead during initial PDF ingestion. Furthermore, vision-language models required for dense diagram captioning require substantial GPU VRAM, presenting deployment hurdles on edge and CPU-only devices.

5. Future Work

We plan to investigate lightweight heuristic diagram classification combined with ONNX-quantized cross-encoders to deliver real-time sub-second query latency on commodity hardware.